Fiscal Year:	FY 2016	**Task Last Updated:**	FY 11/03/2015
PI Name:	Wheeler, Raymond Ph.D.		
Project Title:	Pick and Eat Salad Crop Testing		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SHFH**:Space Human Factors & Habitability (archival in 2017)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	NASA Kennedy Space Center		
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City:	Kennedy Space Center	**State:**	FL
Zip Code:	32899-0001	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	08/26/2014	**End Date:**	12/31/2015
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Massa, Gioia Ph.D. (NASA Kennedy Space Center) Stutte, Gary W. Ph.D. (Vencore-ESC) Spencer, LaShelle E. M.S. (Craig Technologies) Hummerick, Mary E (Vencore-ESC)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NASA is planning for long duration missions beyond low Earth orbit. To maintain crew health and performance, adequate nutrition and acceptability must be provided through the food system to prevent micronutrient deficiency. One approach to address this is to grow fresh foods in situ, focusing on dietary variety and nutrients that are deficient in packaged food, or that diminish over time with storage. In situ food production can reduce consumables and packaging, and provide a path toward self-sufficiency for longer duration missions. Fresh foods (vegetables) could enhance crew diet through expanding nutritional diversity, augmenting flavors, aromas, and an overall food system acceptability. This pilot study will assess the feasibility of growing dwarf tomato and pepper fruits as components of a pick-and-eat supplemental food system for International Space Station (ISS) and future exploration missions. Specific Aims: 1) Identify several cultivars of dwarf tomato and pepper; 2) Establish plants in controlled environment growth chambers and monitor a range of growth and horticultural parameters using flight-like constraints (growing volumes, environmental conditions, watering techniques, and electrical lighting), and nutritional and acceptability attributes; 3) Establish feasibility of growth protocols and cultivar selection for spaceflight, and make recommendations to guide future research solicitations. Research Methods: All plants will be grown in plant growth chambers under fluorescent lighting at an intensity of 300 µmol m-2 s-1, a 16-h photoperiod, 23 degrees C, and elevated CO2 (1500 ppm). These conditions are close to what might be expected in a flight chamber light, such as the Advanced Plant Habitat or Veggie on the ISS. Unlike the leafy vegetables, tomato and pepper fruits will require production cycles of 75-100 days. Plants will be grown to an acceptable harvest point, after which fresh mass, dry mass, and morphological traits will be measured. Tissue samples will be freeze dried for analysis of nutrients such as lycopene, lutein, and anthocyanin using High Performance Liquid Chromatography (HPLC), and elemental analysis using Inductively Coupled Plasma (ICP) spectrometry. In addition, Vitamin K analysis will be performed through a commercial lab. The most promising cultivars will be replanted to produce fruit for acceptance testing at NASA Johnson Space Center (JSC).
Rationale for HRP Directed Research:	This research is directed because it is a pilot study and contains highly constrained research, which requires focused and constrained data gathering and analysis that is more appropriately obtained through a non-competitive proposal.
Research Impact/Earth Benefits:	Testing and selection could help identify shorter growing tomato and pepper cultivars that could have application in terrestrial, vertical agriculture systems.
Task Progress:	Two complete growth cycles with several cultivars of tomatoes and peppers were completed. The different cultivars were evaluated for growth (short height being desirable), yield, and nutritional attributes. Fruit from the second growth cycle were also evaluated for sensory attributes by taste panels. The intent of these studies was to down-select several cultivars for each species (tomato and pepper) as potential candidates for eventual testing in space, such as with the Veggie plant growth unit on the International Space Station.
Bibliography Type:	Description: (Last Updated: 02/08/2019)